pfi

Python for Fixed Income

COURSE 2 · BOND MATH

Spread Over the Curve

WEEK 3 · DAY 2

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The desk lead's note: where does 147 come from?

The Desk Lead's Note

Our month-end file has a column called "oas". Bezane Finance's 2032 bond shows 147. Where does that come from? Rebuild it for all 200 bonds. Then tell me whether our own ten-tenor curve gives the same spreads, and what we should call the number.

The issuers are invented; every number describes the data.

Read the Book

cusip	issuer	maturity	yield_pct	oas
99080CAE8	Bezane Finance	2032-06-18	5.606	147
99049XAE2	Bezamyth Machinery	2029-04-17	9.220	535
99039NAC0	Bezoquil Freight	2030-06-03	5.509	153
99039NAG1	Bezoquil Freight	2053-04-09	6.756	218
99034IAD4	Bezorin Finance	2036-04-05	5.600	127

Read the Curve

```
[0.25, 0.5, 1.0, 2.0, 3.0, 5.0, 7.0, 10.0, 20.0, 30.0]
[3.587, 3.622, 3.69, 3.811, 3.916, 4.086, 4.214, 4.348, 4.54, 4.586]
```

```
................................   [100%]
32 passed in 0.50s
```

One number on one bond first, then the whole book.

12.1 A spread to the curve; 12.2 years the file's way

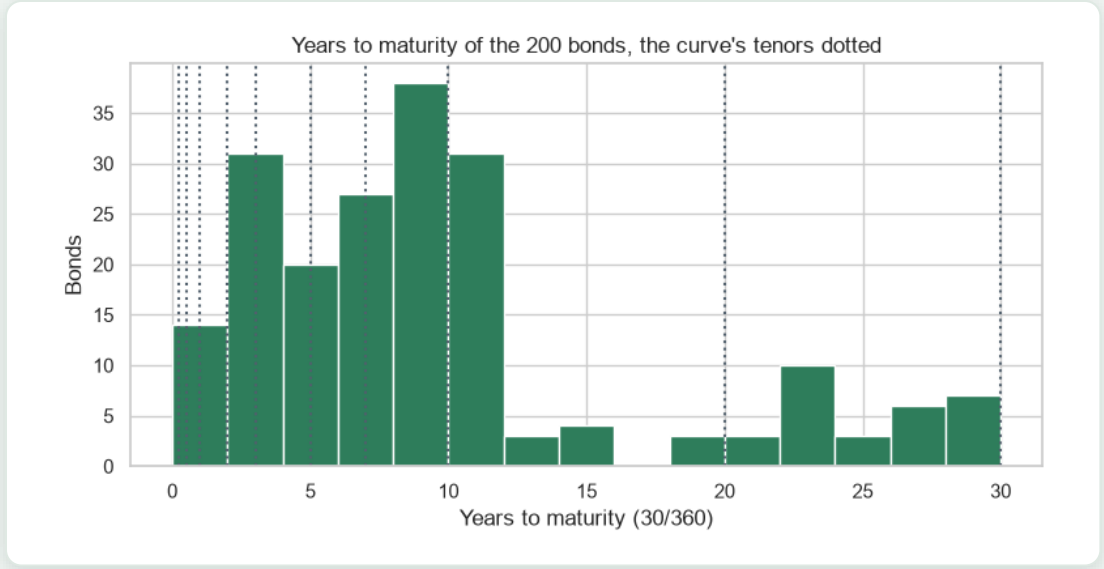
Ten tenors in years, ten yields in percent: notebook 11's invented curve. A spread to the curve is the bond's yield minus the curve's yield at the same maturity. One percent is 100 basis points, so the gap in percent times 100 is the spread in bp.

The bond matures in 2032, between the curve's 5-year point (4.086) and its 7-year point (4.214).

12.2 Years to Maturity, the File's Way

The curve's tenors are years, so each maturity has to become a number of years. The file's own rule, in "tools/make_holdings.py", the script that built it: the 30/360 day count from the as-of date to maturity, over 360.

```
holdings['years'] = [bondmath.days_30_360(a, m) / 360
                     for a, m in zip(holdings['as_of_date'], holdings['maturity'])]
1.0056 to 29.8972 years
```



Every bond sits inside the table, from 1 to 30 years; none falls exactly on a tenor.

Excel side by side. Settlement in D2, maturity in E2; filled down, "YEARFRAC" equals the Python column on all 200 bonds.

Job	Python	Excel
Years, 30/360	`bondmath.days_30_360(as_of, maturity) / 360`	=YEARFRAC(D2,E2,0)
The day count	`bondmath.days_30_360(as_of, maturity)`	=YEARFRAC(D2,E2,0)*360
The lookalike, 28 Feb 2027 to 31 Aug 2031	none	=DAYS360(DATE(2027,2,28),DATE(2031,8,31))

12.3 The curve at Bezane's maturity, and 147

The Straight Line

```
years = holdings.loc[0, 'years']
curve_line_pct = bondmath.interpolate_yield(years, curve_tenors, curve_yields)
print(f'straight line: {curve_line_pct:.6f} percent')
straight line: 4.131867 percent
```

The Formula the File Used

"DATA.md" records it, and notebook 11 met it: 3.55 + 1.05 x (1 - exp(-T / 7)) percent, rounded to 3 decimals, with T the years. The file read its curve another way.

```
import math
curve_formula_pct = round(3.55 + 1.05 * (1 - math.exp(-years / 7)), 3)
```

The Spread, From the Formula

```
print(f"{(bond['yield_pct'] - curve_formula_pct) * 100:.6f} bp; the file says {bond['oas']}")
147.000000 bp; the file says 147
```

147 to the last printed digit.

Excel side by side. Years in G2; the curve's tenors in A2:A11 and yields in B2:B11.

Job	Python	Excel
The straight line	`bondmath.interpolate_yield(years, curve_tenors, curve_yields)`	notebook 11's FORECAST.LINEAR with INDEX and MATCH
The formula	`round(3.55 + 1.05 * (1 - math.exp(-years / 7)), 3)`	=ROUND(3.55+1.05*(1-EXP(-G2/7)),3)

The file read its curve from the formula at the bond's own years, rounded to 3 decimals: 147 to the last digit.

12.4 Two functions, docstring first: years_to_maturity

Three steps make a spread: dates to years, years to a curve yield, the difference times 100. Two functions do them, written the module's way: docstring first, then the code.

```
%%writefile -a bondmath.py
def years_to_maturity(settlement: date, maturity: date) -> float:
    if settlement >= maturity:
        raise ValueError(f"settlement {settlement} must be before maturity {maturity}")
    return days_30_360(settlement, maturity) / 360
```

```
cases = [(date(2026, 9, 30), date(2036, 9, 30)), (date(2026, 9, 30), date(2032, 6, 18)),
         (date(2027, 2, 28), date(2031, 8, 31))]
for start, end in cases:
    print(start, 'to', end, round(bondmath.years_to_maturity(start, end) * 360), 'days')
2026-09-30 to 2036-09-30 3600 days
2026-09-30 to 2032-06-18 2058 days
2027-02-28 to 2031-08-31 1621 days
```

1621: notebook 07's February rule makes the 28 February start the 30th and leaves the 31st alone. That is YEARFRAC's count; "DAYS360" gave 1620.

Dates the Wrong Way Round

```
bondmath.years_to_maturity(date(2032, 6, 18), date(2026, 9, 30))
ValueError: settlement 2032-06-18 must be before maturity 2026-09-30
```

It refuses: a negative number of years would read the curve without complaint.

spread_to_curve_bp, and the whole book

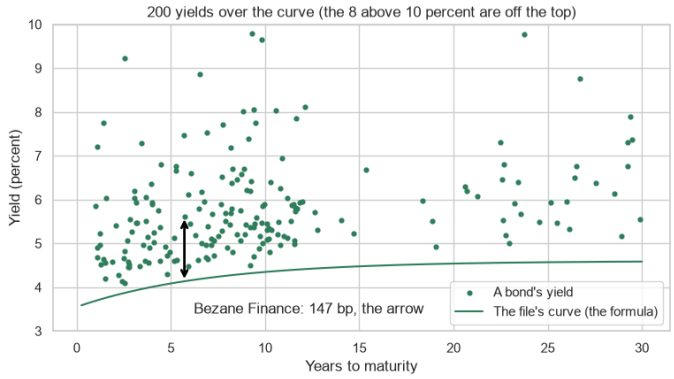
```
%%writefile -a bondmath.py
def spread_to_curve_bp(yield_pct: float, years: float, curve_tenors: list[float],
                       curve_yields_pct: list[float]) -> float:
    curve_yield_pct = interpolate_yield(years, curve_tenors, curve_yields_pct)
    return (yield_pct - curve_yield_pct) * 100
```

The formula is not a table, but the function can still take it. A curve of one point, the bond's own years and the formula's yield there, is flat at that yield everywhere: notebook 11's flat ends.

```
importlib.reload(bondmath)
spread_bp = bondmath.spread_to_curve_bp(bond['yield_pct'], years, [years], [curve_formula_pct])
147.000000 bp
```

```
holdings['curve_formula_pct'] = [round(3.55 + 1.05 * (1 - math.exp(-t / 7)), 3)
                                 for t in holdings['years']]
holdings['spread_formula_bp'] = [
    bondmath.spread_to_curve_bp(y, t, [t], [c])
    for y, t, c in zip(holdings['yield_pct'], holdings['years'], holdings['curve_formula_pct'])]
largest_miss = (holdings['spread_formula_bp'] - holdings['oas']).abs().max()
assert largest_miss < 1e-9, "a bond's spread does not reproduce its oas"
largest miss: 2.3e-13 bp
```

2.3e-13 bp is float rounding, as in Course 1's 0.1 + 0.2: zero for any desk.



Each bond's height above the line is its spread.

Excel side by side. The yield in F2 is typed in percent, as in the file, and no Excel function receives it, so there is no "/100".

```
=(F2-J2)*100
146.99999999999997
```

Two functions rebuild the "oas" column for all 200 bonds, to 2.3e-13 of a basis point.

12.5 Tests from Excel

Each case is a 30/360 count Excel's "YEARFRAC(settlement, maturity, 0)" x 360 returned for the same dates (the course's Excel run of 4 October 2026).

```
%%writefile -a test_bondmath.py
import math
def test_years_to_maturity_cases():
    cases = [
        (date(2026, 9, 30), date(2036, 9, 30), 3600),  # exactly ten years
        (date(2026, 9, 30), date(2032, 6, 18), 2058),  # the first bond in holdings.csv
        (date(2027, 2, 28), date(2031, 8, 31), 1621),  # the 31st end stays 31 after a 28 February start
    ]
    for settlement, maturity, days in cases:
        assert bondmath.years_to_maturity(settlement, maturity) == pytest.approx(days / 360, abs=1e-12)
```

Six years is halfway from 5 to 7, so the curve is the average of 4.086 and 4.214, 4.150, and a 5 percent yield is 85.0 bp over it.

```
%%writefile -a test_bondmath.py
def test_spread_to_curve_known_case():
    curve_at_6_years = 4.086 + (4.214 - 4.086) * (6 - 5) / (7 - 5)
    expected_bp = (5.0 - curve_at_6_years) * 100
    result = bondmath.spread_to_curve_bp(5.0, 6.0, CURVE_TENORS, CURVE_YIELDS_PCT)
    assert result == pytest.approx(expected_bp, abs=1e-9)
```

Five bonds typed in with their CUSIPs; a failing row names its CUSIP.

```
%%writefile -a test_bondmath.py
def test_spread_reproduces_oas_rows():
    rows = [
        ("99080CAE8", date(2032, 6, 18), 5.606, 147),
        ("99049XAE2", date(2029, 4, 17), 9.22, 535),
        ("99039NAC0", date(2030, 6, 3), 5.509, 153),
        ("99039NAG1", date(2053, 4, 9), 6.756, 218),
        ("99034IAD4", date(2036, 4, 5), 5.6, 127),
    ]
    for cusip, maturity, yield_pct, oas in rows:
        years = bondmath.years_to_maturity(date(2026, 9, 30), maturity)
        curve_yield_pct = round(3.55 + 1.05 * (1 - math.exp(-years / 7)), 3)
        result = bondmath.spread_to_curve_bp(yield_pct, years, [years], [curve_yield_pct])
        assert result == pytest.approx(oas, abs=1e-9), cusip
```

```
.................................   [100%]
35 passed in 0.50s
```

32 from notebooks 01 to 11, and today's three.

Commit, and when every test arrived

```
!git -C "{work}" diff --stat
!git -C "{work}" add bondmath.py test_bondmath.py
!git -C "{work}" commit -q -m "Notebook 12: years_to_maturity and spread_to_curve_bp"
 bondmath.py    | 27 +++++++++++++++++++++++++
 test_bondmath.py | 40 ++++++++++++++++++++++++++++++++++++++++
 2 files changed, 67 insertions(+)
```

As in notebook 09, "--" and a file name keep only the commits that changed that file; notebook 05's "--decorate" shows the tags.

```
!git -C "{work}" diff --stat week2
 bondmath.py    | 57 +++++++++++++++++++++++++++++++++++++++++++++++++++++
 test_bondmath.py | 69 ++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++
 2 files changed, 126 insertions(+)
```

"git diff --stat" with a tag compares that commit with the files now. Notebooks 11 and 12, the first two of week 3.

```
!git -C "{work}" log --oneline --decorate -- test_bondmath.py
4c7dcbc (HEAD -> main) Notebook 12: years_to_maturity and spread_to_curve_bp
10f4dad Notebook 11: interpolate_yield
90faa12 (tag: week2) Notebook 10: convexity
5fcb7c6 Notebook 09: macaulay_duration, modified_duration, and dv01
20b3546 Notebook 08: price, dirty_price, and yield_to_maturity
98124d3 Notebook 07: days_30_360, coupon_period_days, and accrued_interest
9fe8f04 Notebook 06: add_months, coupon_dates, and a .gitignore
322960d (tag: week1) Notebook 05: tests of par pricing and of price against yield
7147a11 Notebook 04: yield_from_price
3f57cac Notebook 03: convert_yield
408a678 Notebook 02: cash_flows and price_from_yield
d029474 Notebook 01: future_value and present_value
```

All twelve: every notebook added tests. The same command with "bondmath.py" leaves out notebook 05, which added tests only.

35 tests pass, and the history says when every test arrived.

12.6 The desk's ten-tenor table, and the gap

Every Spread on the Table

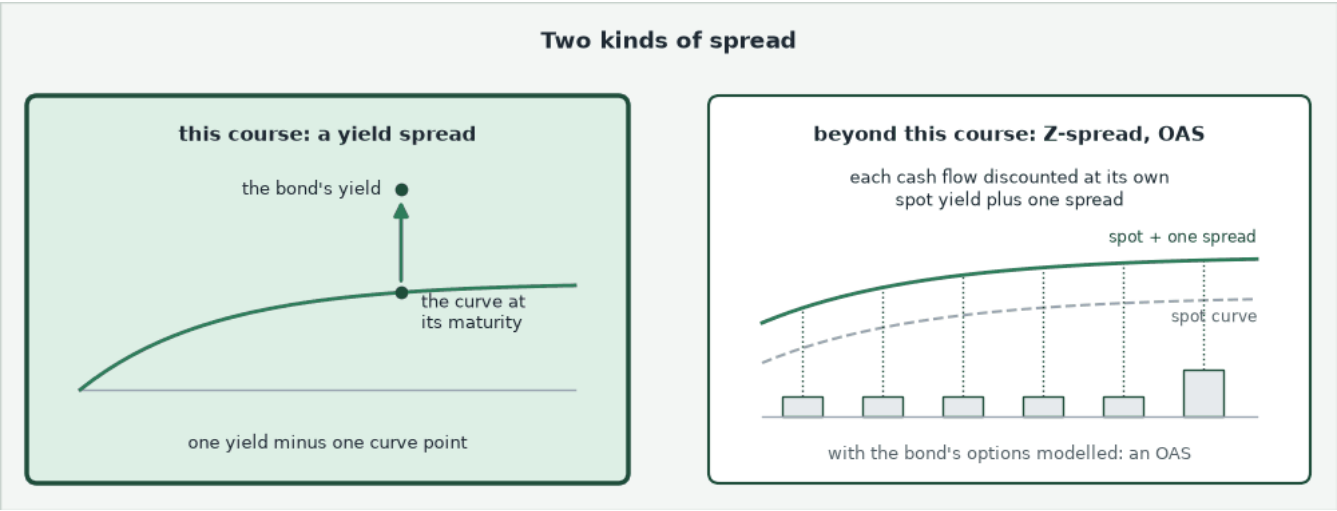
```
holdings['curve_table_pct'] = [bondmath.interpolate_yield(t, curve_tenors, curve_yields)
                               for t in holdings['years']]
holdings['spread_table_bp'] = [bondmath.spread_to_curve_bp(y, t, curve_tenors, curve_yields)
                               for y, t in zip(holdings['yield_pct'], holdings['years'])]
holdings['gap_bp'] = holdings['spread_table_bp'] - holdings['oas']
print(f"Bezane Finance on the table: {holdings.loc[0, 'spread_table_bp']:.2f} bp")
Bezane Finance on the table: 147.41 bp
```

Excel side by side. The straight line in H2, from 12.3's table.

Job	Python	Excel	Result
Spread to the table, bp	`bondmath.spread_to_curve_bp(y, t, curve_tenors, curve_yields)`	=(F2-H2)*100	147.41333333333327

The same bond on the same curve is 147 or 147.41: compare two spreads only when they share the curve, the reading between tenors, and the count of years.

12.7 What to call the number



The file calls its column "oas", short for option-adjusted spread. The file built it as a yield spread: the bond's yield less the curve's yield at the same maturity. Measured over a government curve, desks call this a G-spread. A desk's OAS adds one spread to a spot curve, a zero-coupon yield for each date, and discounts every cash flow on it, with the bond's options valued by a model of moving rates. For a bullet with no options, the OAS is close to the Z-spread: the same one spread, with no options to value. Spot curves and Z-spreads are beyond this course.

The course calls it "spread_to_curve_bp", a yield spread at the bond's own maturity; the file keeps the name "oas".

Bring it home, and what you can now do

Copy today's two files into your own "my_bondmath" folder, then test and commit there. The copy replaces your files: anything of your own must be in the work folder first.

```
Set-Location "$HOME\projects\my_bondmath"
$work = "$env:TEMP\pfi_course2_12_k4w7q2xr"
Copy-Item "$work\bondmath.py", "$work\test_bondmath.py", "$work\.gitignore" .
python -m pytest -q
git diff --stat
git add bondmath.py test_bondmath.py
git commit -m "Notebook 12: years_to_maturity and spread_to_curve_bp"
git log --oneline -- test_bondmath.py
git diff --stat week2
```

If "week2" is unknown, tag notebook 10's commit: "git tag week2" followed by its hash. In Colab, download the two files from the work folder in the file browser. The work folder stays in your temp folder.

You Can Now

Say what a spread to the curve measures, in bp.

Count years to maturity on 30/360 with "years_to_maturity", as Excel's "YEARFRAC" with basis 0 does.

Compute a spread with "spread_to_curve_bp" on any curve, a one-point curve included.

Rebuild the file's "oas" column from the curve's formula.

Measure how far the reading between tenors moves a spread, and say when two spreads compare.

Call the number a yield spread, and list the commits that changed one file.

Predict first: the curve yield, and two counts

Predict 1

Bezane Finance yields 5.606 percent and its "oas" says 147. What curve yield, in percent, did the file subtract? Write down a number.

```
bond = holdings.iloc[0]
implied_curve_pct = bond['yield_pct'] - bond['oas'] / 100
print(f"{bond['issuer']}: the file subtracted {implied_curve_pct:.3f} percent")
```

Predict 2

Count Bezane Finance's years as actual days over 365, then as 30/360 days over 360. How far apart are the two answers?

A) about a month: two year lengths five days apart, for more than five years

B) identical: the same two dates give the same years

C) a day or two: each count is divided by its own year length

```
as_of, maturity = bond['as_of_date'], bond['maturity']
actual_days = (maturity - as_of).days
days_360 = bondmath.days_30_360(as_of, maturity)
print(f'actual: {actual_days} days / 365 = {actual_days / 365:.4f} years')
print(f'30/360: {days_360} days / 360 = {days_360 / 360:.4f} years')
```

Predict first: the gaps, and the other count

Predict 3

Notebook 11 averaged the line's distance below the formula over an even grid from 0.25 to 30 years: 1.01 bp. Averaged over the 200 bonds, at their own maturities, the size of the gap is:

A) smaller: most bonds mature where the tenors sit close together

B) about the same: the same curve and the same straight lines

C) larger: the file's rounding adds to every gap

```
size = holdings['gap_bp'].abs()
print(f'average {size.mean():.2f} bp, largest {size.max():.2f} bp')
print(f"under 10 years: {(holdings['years'] < 10).sum()} of 200 bonds")
print(f"table spread larger: {(holdings['gap_bp'] > 0).sum()} of 200 bonds")
holdings.loc[size.idxmax(), ['cusip', 'issuer', 'years', 'curve_formula_pct', 'curve_table_pct',
                             'gap_bp']]
```

Predict 4

Rerun the formula, rounding and all, on actual days over 365. The largest miss against "oas" is:

A) thousandths of a bp: the years barely move

B) about a tenth of a bp: the rounded curve moves in whole steps or not at all

C) about 3 bp, the size of the table's gaps

```
actual_years = [(m - a).days / 365 for a, m in zip(holdings['as_of_date'], holdings['maturity'])]
actual_curve = pd.Series([round(3.55 + 1.05 * (1 - math.exp(-t / 7)), 3) for t in actual_years])
miss = ((holdings['yield_pct'] - actual_curve) * 100 - holdings['oas']).abs()
print(f'largest miss {miss.max():.3f} bp; {(miss > 1e-9).sum()} of 200 bonds miss')
```

The note, answered

The note asked	Answer
All 200 rebuilt	to 2.3e-13 bp, two functions, 35 tests
What to call it	a yield spread to the curve (a G-spread), not an OAS

Be proud: you can say exactly how a spread on someone else's file was built.

Next is Spread Duration and DTS: what the price does when the spread moves.